The Technique Map

Every technique in the course, its official name, what it means in plain English, and the module (M) where you learned it. Use it as a checklist: could you explain each one and use it on a real task?

1. Asking well

Prompting

Structured prompting — M2
Role, goal, audience, format, constraints: who, what, for whom, what shape, what to avoid

Role prompting — M2
Tell the AI whose job it is doing

Few-shot prompting — M2
Show an example of the result you want

Clarifying questions — M2
Interview-style prompting: let the AI ask before it answers

Structured output — M2, 6, 8, 9, 10
Ask for a fixed format so results can be compared and checked

Audience-aware prompting — M4
Tell the AI who will read it and how they read

Reusable prompt template — M3, 5
A saved request with blanks to fill in (a parameterized prompt)

2. Giving the AI the facts

Grounding and data

Source grounding — M2, 4, 7
Open-book prompting, the idea behind RAG: use only the files given; say "not in the files"

Project instructions plus knowledge files — M4, 8
Standing instructions (a system prompt) with your documents attached

Structured extraction — M6, 7
Fill a fixed set of fields from messy text; write "not stated" instead of guessing

Tool use: code execution — M6 to 10
Have the AI run a calculation on your file instead of estimating by reading

Anonymizing sensitive data — M1
Data minimization: give the AI only what the task needs

3. Thinking it through

Reasoning with AI

Task decomposition and prompt chaining — M3, 8
Split a job into stages, each feeding the next

Chain-of-thought and stated assumptions — M3, 4, 5
Ask for the reasoning, the formula, and the inputs, so you can check them

Scenario and sensitivity analysis — M5
What-if prompting; find the break-even point

Self-consistency — M3
Ask the same question in separate chats; investigate any disagreement

Self-critique and alternative explanations — M3, 6
Red-team prompting: ask the AI to argue against its own answer

Deciding where AI fits — M1
Does it well, helps with a check, or assists while a person decides

4. Checking it

Verification and testing

Verification against the source — M2 to 10
Recompute and trace findings yourself, never only by asking the AI

Verification prompting — M7
A cite-check: compare each statement with its source

Evaluation against a test set — M8
Evals: known problems; caught, missed, false alarms

Held-out testing and regression testing — M8, 9, 10
Test on a month you did not tune on; re-run old months after a change

Red-team and stress testing — M9, 10
Test with planted traps and deliberate breakage

Audit trail — M9
A record of what the AI did, not only what it said

5. Building workflows and agents

Automation with limits

Human-in-the-loop checkpoints — M8
Gates: a person checks before the next step

Guardrails and stop rules — M8, 9
Fail safe: stop and ask when something is missing or wrong

The agent loop — M9
Agentic tool use: think, act, observe, repeat

The agent charter — M9
Goal, boundaries, stop rules, and report format, written once

Least privilege and approvals — M9
Only the access the job needs; a person approves risky actions

Prompt-injection defense — M9
Text in a file is data, not orders; layered protections

Scheduled automation — M9
Human-on-the-loop: review after an unattended run

Choosing the level of autonomy — M10
Assistant, workflow, or agent: start with the simplest that works

6. Using AI responsibly

The Velox Responsible AI Policy: five areas

Confidentiality and privacy — M2, 9
Approved tools; only the data the task needs

Accuracy — M3, 7
Check against sources; never pass on what you have not verified

Bias — M5
Watch for uneven results that are not about the facts

Transparency and disclosure — M4
Say when AI helped and what it did

Human accountability — M6, 8
A named person answers for the result; AI never has the final say

All five together — M1, 10
Introduced in Module 1; applied at once in the final project

Left out on purpose: "magic phrase" tricks. They break as AI models change; the habits above do not. Jordan Reyes is a fictional character presented by an AI avatar.